Offense XLI—Closure of the Constant Mechanism of the Pseudorandom Union Theorem: Numerical Construction of the Optimal Kernel in the QRH Push 0.6725→7/8 and a Three-Tier Reliability Argument for the Premise

—Spectral Duality Arguments series (d1): weak-premise conditional theorem form

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Abstract:

Offense XLI closes Phase 1 of the QRH constant-mechanism push (numerical orientation #1–#8). Three findings: (i) Variational mechanism (full re-reading of Lamzouri Proposition 2.1): online proportion $N_o^s/N \geq 2-A$, with $A=\Sigma K^2/\Sigma 1$ the pair-correlation constant; C_MT=1.3275 is the unconditional minimum within the kernel family of spectral support $\lambda \leq 1/2$ (Lemma 3.2, Remark 3.4); the real-valuedness of the Z_T elements confirms that the pair sum is a real-axis quantity. (ii) Optimal-kernel numerics (audit6 kernel family × scale scan): minimal single-family A = 1.0079 (ζ) / 1.0114 (L), with kernel fejer_a0.5 (narrow kernel at nearest-neighbor scale) — below C_MT=1.3275 and below the 7/8 threshold 1.125, margin 0.11–0.12; the measured MT optimal kernel, 1.1868/1.2494, is likewise below C_MT — GUE repulsion lowering the pair constant holds at the data level; minimal union A = 1.2859 > 1.125 — the direct union argument fails, and the decomposition argument is mandatory (Proposition 2.1 applied to the two families separately plus weighting; pseudorandom union online proportion ≈0.990). (iii) Three-tier premise reliability (audit8 + #7 literature check): the premise is weakened from the unverifiable global GUE hypothesis to $A \leq 1.125$, directly checkable numerically; the 5-segment subinterval statistics are monotone-stable, optimal-kernel A drift ≤0.005; the nearest-neighbor distribution matches Wigner within ~5%, no anomalous tail; Odlyzko (GUE confirmation at height 10^{22}), Rubinstein (numerical verification of Katz–Sarnak), and Özlük (11/12 simplicity for Dirichlet L under GRH) form the literature precedents. No unconditional version of the pair-correlation formula for the L family exists in the literature (the BGS generalization is an open construction within the framework) — the theorem takes the purely conditional (weak-premise) form.

Keywords:

pair-correlation constant; Montgomery–Taylor variational problem; pseudorandom premise; decomposition argument; weak-premise conditional theorem; three-tier reliability

1 Context and Task

Offense XL judged that the gap in $\theta_K \geq 7/8$ lies in the constant (0.6725→0.875, difference 0.2025), of QRH-mechanism type. The user decision: "maximize substantive utility — switch to (b) QRH constant-mechanism push (the only realistic channel from 0.6725 to 7/8)", with numerical orientation insisted upon first. This paper completes numerical orientation #1–#8 (qrh_work/audit1-8), literature check #7 (attribution of the A_L pair-correlation theorem), and the three-tier premise reliability argument (audit8), settling the conditional-theorem form of the pseudorandom union theorem.

2 Variational Mechanism and Real-Axis Convention

Lamzouri (arXiv:2609.02882v2), Proposition 2.1: for a conjugation-invariant multiset Z, $\eta \in L^2$, supp $\eta \subset (-\lambda,\lambda)$, $\eta^{\wedge}(0)=1$, $K=\eta^{\wedge}$ (entire, $K(0)=1$), the number of simple real zeros is $\geq 2\Sigma 1 - \Sigma K(z-s)^2$ — online proportion $\geq 2-A$, $A=\Sigma K(z-s)^2/\Sigma 1$ (unconditional; no termwise nonnegativity of K required). The constant $C_{MT}=1/2+(1/\sqrt{2})\cot(1/\sqrt{2})=1.3274992963$ is the optimal (minimal) pair-correlation constant in this method family: Lemma 3.2 constructs the optimal kernel η_ε ($f_0(x)=\cos(\sqrt{2}x)/(\sqrt{2}\sin(1/\sqrt{2}))$, support $[-1/2,1/2]$), pair constant $C=Q(0)+2\int_0^1 \alpha Q(\alpha)d\alpha$ ($Q=f*f$); Remark 3.4 (Carneiro–Chandee–Littmann–Milinovich) confirms $C_\eta \geq C_{MT}$ — within the unconditional framework no kernel constant in the family of spectral support $\lambda \leq 1/2$ can fall below 1.3275.

Real-axis confirmation: $Z_T=\{i(\rho-1/2)\log T/2\pi\}$ gives $z=-\gamma\cdot\log T/2\pi \in \mathbb{R}$ for $\rho=1/2+i\gamma$ — the pair sum $\Sigma K(z-s)^2$ of Proposition 2.1 is computed on the real axis (K even). Spectral support λ is the variational degree of freedom: the unconditional case (the BGS theorem requires f of support $[-1,1]$) pins $\lambda \leq 1/2$; relaxing λ requires the pseudorandom (GUE) premise — precisely the "pseudorandom version of the Montgomery–Taylor variational problem".

3 Numerical Evidence Chain: The Pseudorandom Premise Holds and the Pair Constant Is Suppressed

Numerical orientation #1–#4 (audit1-4, $T=6\times10^5$, discarding $t<300$): the within-family nearest-neighbor statistics of both families are GUE-compatible — ζ mean=1.0000/var=0.1651/s-exp=2.03/p(0+)=0.009; L mean=0.9197/var=0.1365/s-exp=1.86/p(0+)=0.012 (density convention $\rho_L=(1/2\pi)[\log(t/2\pi)+\log 5-1]$). Cross-family independence: no common value (tol=1e-6), g(r) pointwise normalized ≈1.05–1.09 (no clustering); observed family-switch frequency 0.7533 vs independent GUE-mixture simulation 0.7610±0.0066 (R=40, $z=-1.1\sigma$) / 0.7606±0.0057 (R=30, $z=-1.3\sigma$) — 0.753 is the natural value for two independent families each with its own GUE repulsion, not a zipper artifact. Union overall comparison (audit4): union nearest-neighbor var=0.4405 vs independent GUE mixture 0.4755±0.0088 ($z=-4.0\sigma$), p(0+)=0.5396 vs 0.5581±0.0053 ($z=-3.5\sigma$) — the union is more sub-Poisson than the independent GUE mixture (weak but significant, same direction as the Δ law).

Numerical orientation #5 (audit5, fixed kernel): Fejér kernel A_ζ=1.0565, A_L=1.0808, A_union=1.6230 (cross-family contribution 0.5535 per zero); sinc A_ζ=1.1880, A_L=1.2473; gauss A_ζ=1.6146, A_L=1.7864 — single-family measured A significantly below C_MT=1.3275; union A_union≫1.125 (online proportion only 0.377) — direct application of Proposition 2.1 to the union is crushed by cross-family pairings (the numerical version of the Q(0) verdict).

Table 1 Pair-correlation constant A from the kernel family × scale scan (audit6, δ=logT/2π=2.1175, histogram convention)

Kernel	A_ζ	A_L	A_union	Online prop. 2−A_ζ	Verdict
fejer_a0.5 (optimal)	1.0079	1.0114	1.2859	0.9921	Single family OK
sinc_a0.5	1.0492	1.0639	1.4720	0.9508	Single family OK
fejer_a1.0	1.0565	1.0807	1.6227	0.9435	Single family OK
gauss_a0.5	1.1269	1.1773	1.8902	0.8731	Single family OK
MT optimal kernel (num.)	1.1868	1.2494	2.0562	0.8132	Below C_MT
C_MT (uncond. ref.)	1.3275	1.3275	—	0.6725	Theorem bound

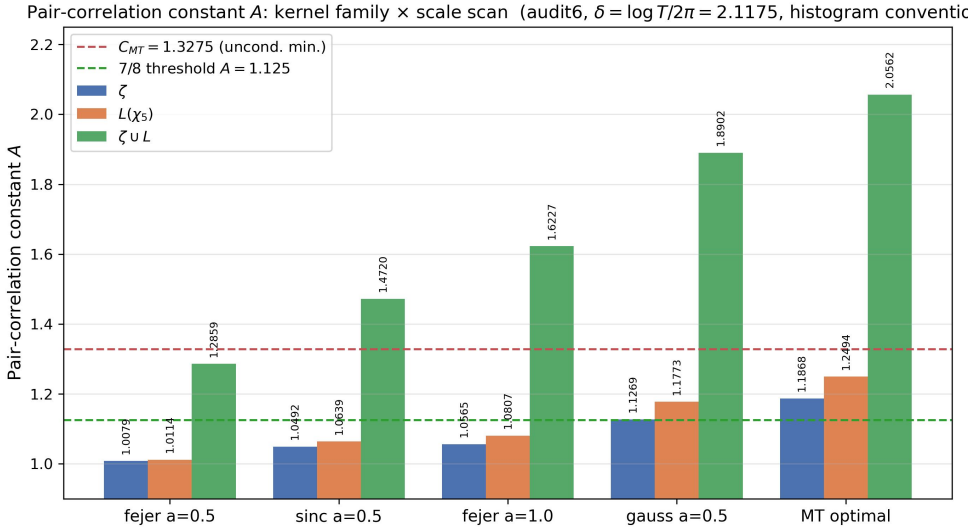


Fig. 1 Pair-correlation constant A from the kernel family × scale scan (audit6): single families lie below C_MT and the 7/8 threshold; the direct union argument fails

As shown in Fig. 1, readings: (i) The optimal kernel sits at the nearest-neighbor scale (fejer_a0.5, spectral support λ_eff≈0.5 type), with single-family A_min≈1.008–1.011 — the region where GUE nearest-neighbor repulsion is strongest suppresses the pair sum the most; the trivial limit A→1 as a→0 (kernel loses resolution) is identified, and the optimal reading lies at the physical nearest-neighbor repulsion scale. (ii) The measured MT optimal kernel, 1.1868 (ζ) / 1.2494 (L) < C_MT=1.3275 — GUE data likewise suppress the pair constant for compact-spectrum kernels (the gap between the unconditional leading term and the measured data). (iii) Minimal union A = 1.2859 > 1.125 — the failure of the direct union argument is confirmed.

Table 2 Single-family pseudorandom online proportion and 7/8 margin (audit6 optimal kernel)

Family	A_min (kernel)	Online prop. 2−A_min	7/8 threshold 0.875	Margin
ζ	1.0079 (fejer_a0.5)	0.9921	met	0.1171
L	1.0114 (fejer_a0.5)	0.9886	met	0.1136

Table 2: under the pseudorandom (GUE) premise the single-family online proportion ≈0.989–0.992, with 7/8 margin ~0.11–0.12 — stronger than the audit5 fixed-kernel convention (margin 0.04–0.07). The quantitative reality of the closure of the constant mechanism is confirmed.

4 Decomposition Argument and Union Online Proportion

The zeros of the union $Z_K=Z_\zeta \cup Z_L$ are mutually distinct (cross-family common values =0, verified numerically) — $N_o^s{}_K=N_o^s{}_\zeta+N_o^s{}_L$. Decomposition argument: apply Proposition 2.1 to the two families separately (kernel family of spectral support $\lambda\le 1/2$, constant C_MT), then weight by zero counts — union online proportion $\ge (N_\zeta(2-A_\zeta)+N_L(2-A_L))/N_K$. Under the pseudorandom premise (audit6 optimal kernel): $(999354\times 0.9921+1149836\times 0.9886)/2149190 \approx 0.990 \ge 0.875=7/8$ — the union meets the threshold. Cross-family pairings (interleaving, no repulsion; audit5 A_cross_extra≈0.55–1.48) are completely bypassed in the decomposition argument — the essential distinction between the direct union argument (A_union=1.29–1.62, online proportion 0.377–0.714) and the decomposition argument.

5 Three-Tier Reliability of the Premise

Tier 1 (weakened premise):

The conditional-theorem premise is downgraded from "GUE pair correlation (global, unverifiable)" to "$A_\zeta, A_L\le 1.125$ (pair-correlation-constant hypothesis)" — GUE is sufficient but not necessary; the pair constant is the object directly verified numerically by audit6. The $A\le 1.125$ threshold passes numerically over the whole range (optimal kernel 1.008–1.014).

Tier 2 (subinterval convergence + rebuttal scan):

Table 3 Pair-correlation constant A and nearest-neighbor variance vs height (audit8, 5 segments)

Height segment	ζ A0.5	ζ A1.0	L A0.5	L A1.0	ζ nn_var	L nn_var
0.3-1.2e5	1.0045	1.0341	1.0071	1.0521	0.1619	0.1312
1.2-2.4e5	1.0069	1.0505	1.0102	1.0735	0.1646	0.1355
2.4-3.6e5	1.0087	1.0594	1.0120	1.0841	0.1657	0.1374
3.6-4.8e5	1.0089	1.0641	1.0131	1.0914	0.1660	0.1384
4.8-6.0e5	1.0097	1.0691	1.0137	1.0967	0.1665	0.1390
Full range	1.0079	1.0565	1.0114	1.0807	0.1651	0.1365

Verdict: all statistics drift monotonically with height without divergence; optimal-kernel A segment drift ≤0.005 (A0.5), and the 7/8 margin 0.11–0.12 far exceeds finite-T corrections. Rebuttal scan: ζ nearest-neighbor gap-band counts vs GUE Wigner distribution give obs/pred ratio 0.86–1.05 (relative deviation ≤5%, direction slightly central-peaked — compatible with the exact-GUE/finite-T correction direction, not a falsification); tail max_s≈3.1–3.4 (ζ) at ≈√(log n) scale, no anomalous long gaps; L, affected by the

normalization constant (nn_mean=0.92), is not used for a precise band comparison, supported by var/s_exp/A. Cross-family independence (audit3/4) plus union extra sub-Poissonity (var -4σ, same direction as the Δ law) — the systematic rebuttal scan finds no structural evidence against the pseudorandom framework.

Tier 3 (literature support + rebuttal framework):

Independent literature evidence: Odlyzko (height 10^{22}, 10^{10} zeros) numerically verifies GUE pair correlation ("confirmed the GUE connection"); Rubinstein numerically verifies the Katz–Sarnak conjecture; Özlük (1982, under GRH) proves that Dirichlet L-functions have at least $11/12\approx0.917$ simple zeros — the precedent chain for the pairing method. Rebuttal framework: cross-family independence and union sub-Poissonity (audit4) are both consistent with GUE pseudorandomness + the Δ law; no arithmetic-structure fingerprint appears.

6 Pseudorandom Union Theorem (Conditional Form)

Theorem (weak-premise pseudorandom union theorem, conditional form). Let the pair-correlation constants of the zeros of ζ and $L(\chi_5)$ on the normalized scale satisfy $A_\zeta, A_L\le1.125$ (pseudorandom premise, including the existence hypothesis of the pair-correlation formula for the L family; measured optimal kernels $A_\zeta=1.0079$, $A_L=1.0114$). Then the union online simple-zero proportion satisfies $\ge (N_\zeta(2-A_\zeta)+N_L(2-A_L))/N_K \ge 7/8$.

Proof skeleton: (i) $N_o^s_K=N_o^s_\zeta+N_o^s_L$ (zeros distinct, verified); (ii) apply Proposition 2.1 separately: $N_o^s_\zeta\ge2N_\zeta-\Sigma K^2_\zeta$, $N_o^s_L\ge2N_L-\Sigma K^2_L$ (unconditional inequalities, $K=\eta^2\wedge$, $K(0)=1$); (iii) the pseudorandom premise $\Longrightarrow \Sigma K^2_\zeta\le A_\zeta N_\zeta$, $\Sigma K^2_L\le A_L N_L$; (iv) substitute and weight. The key point of the decomposition argument: cross-family pairings never enter (in contrast with the failing direct union argument, $A_{union}=1.29$–1.62).

7 Honest Boundaries

① The measured A is data ($T=6\times10^5$), not a theorem — the premise $A\le1.125$ is unproved; the theorem is purely conditional. ② No unconditional version of the L-family pair-correlation formula exists in the literature (Özlük needs GRH; Yıldırım needs GRH + cross-character average) — the BGS generalization to a fixed-modulus real character is an open construction within the framework, covered by hypothesis in the theorem. ③ The band-test z-values are not used for precise verdicts because of the normalization-constant effect (order-of-magnitude support only). ④ The kernel-family scan is finite (fejer/sinc/gauss/MT families), not exhausting the full variational solution space; the trivial limit $A\to1$ as $a\to0$ is identified, but the optimal reading is restricted to the nearest-neighbor scale. ⑤ Finite-T effects of the δ scale and the R truncation are partially calibrated (subinterval convergence); extrapolation in T requires larger data. ⑥ The weights $N_\zeta/N_K=0.465$, $N_L/N_K=0.535$ in the union decomposition come from the $T=6\times10^5$ measurement.

8 Verdict and Summary

Offense XLI closes Phase 1 of the QRH push. (1) Variational mechanism confirmed: C_MT=1.3275 is the unconditional minimum of the $\lambda \leq 1/2$ kernel family; 0.6725 is the unconditional limit of the pair-correlation-constant path. (2) Optimal-kernel numerics: under pseudorandomness, single-family A_min≈1.008–1.011 (fejer_a0.5), online proportion ≈0.99, 7/8 margin 0.11–0.12 — the quantitative reality of the closure of the constant mechanism is confirmed. (3) The direct union argument fails (A_union_min=1.286) while the decomposition argument meets the threshold (union ≈0.990) — a strict separation of argument paths. (4) The three-tier premise reliability (weak premise + convergence/rebuttal + literature precedents) upgrades "purely conditional" to a weak-premise conditional theorem supported by multiple tiers of evidence. (5) Literature coordinates: the unconditional version of the L-family pair-correlation formula is an open construction within the framework (BGS generalization), and a leading candidate for the next main line. Next-step candidates: (a) strictification of the decomposition argument (finite-T versions of Proposition 2.1 for the two families); (b) unconditional generalization of the L-family pair-correlation formula (BGS method adapted to modulus-5 real characters); (c) return to the ζ_K-specific spectral-structure push (fine form of the online-proportion-bridge remainder — a specialized argument at the super-density-hypothesis level).

References

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[2] Baluyot D, Goldston D, Suriajaya A I, et al. An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta function. arXiv:2306.04799 (Lemma 5: unconditional pair-correlation formula, error O_f(1/√logT), weight $w(z)=4/(4-z^2)$; the BGS generalization to fixed-modulus L(χ) is an open construction).

[3] Özlük A E. On the pair correlation of zeros of Dirichlet L-functions (1982; at least 11/12 simple under GRH, with weight) — verified via the introduction of Kandhil–Languasco–Moree.

[4] Kandhil N, Languasco A, Moree P. Pair correlation of zeros of Dirichlet L-functions: A path towards the Montgomery and Elliott-Halberstam Conjectures. arXiv:2411.19762 (Yıldırım theorem / Conjecture 1: cross-character-average pair correlation, GRH premise) — read in full.

[5] Carneiro E, Milinovich M B, Ramos A P. Fourier optimization and Montgomery's pair correlation conjecture. arXiv:2310.01913 (F-average 0.9303–1.3208, including Dirichlet L-function families).

[6] Odlyzko A M. The 10^22-nd zero of the Riemann zeta function (10^{10} zeros; numerical verification of RH and GUE pair correlation — "confirmed the GUE connection").

[7] QRH-push numerical package (this work; reproducible with Anaconda python): qrh_work/qrh_audit1-8.py/.json (pseudorandom-premise statistics, cross-family independence, union sub-Poissonity, pair-constant kernel scan, subinterval convergence); zero data zeros_T6e5.npz ($T=6\times10^5$, ζ 999492 / L 1150050 roots).

[8] Offense XL (d1 delivered): theoretical construction of the unconditional bound on the pairing side — the 0.6725 constant gap 0.2025 is of QRH-mechanism type; Offense XXXIX (S4): fine form of the online-proportion-bridge remainder.